Solution Requirements

2. Requirement Analysis | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
Credit Card Approval Prediction	Ankitha Vana	Finance	July 1, 2026

Functional Requirements

ID	Requirement	Priority
FR-01	Load applicant demographic, financial, and credit payment history datasets.	High
FR-02	Remove duplicate applicant records and handle missing values.	High
FR-03	Merge applicant data with credit history using applicant identifiers.	High
FR-04	Convert credit status records into binary approval/risk labels.	High
FR-05	Train Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers.	High
FR-06	Compare models using accuracy, precision, recall, F1-score, and confusion matrix.	High
FR-07	Save the best-performing model and preprocessing objects.	High
FR-08	Provide a Flask web page for input and prediction output.	High
FR-09	Support optional IBM Watson Machine Learning deployment.	Medium

Non-Functional Requirements

- Prediction response should be fast enough for real-time form submission.
- The interface should be simple and readable for non-technical users.
- Code should be modular so training, preprocessing, and Flask routes are easy to maintain.
- Model output should support compliance review and future audit logging.
- The deployment design should allow scaling to cloud infrastructure.

System Requirements

Category	Requirement
Hardware	Intel i3 or above, minimum 4 GB RAM, recommended 8 GB RAM, and at least 2 GB free storage.
Internet	Required for dataset download, package installation, GitHub access, and cloud deployment.
Operating System	Windows, Linux, or macOS.
Software	Python 3.8+, Anaconda Navigator, Jupyter Notebook, PyCharm or VS Code, and a modern browser.